# Qatar-2b

R-0946

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_180319` · created 2026-07-29T13:47:59+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2458196.8804 +/- 0.0006 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.205 +/- 0.044 |
| Transit depth (Rp/Rs)^2 | 4.19 +/- 1.81 [%] |
| Orbital Inclination (inc) | 86.79 +/- 1.64 |
| Airmass coefficient 1 (a1) | 1.17 +/- 0.043 |
| Airmass coefficient 2 (a2) | -0.105 +/- 0.092 |
| Scatter in the residuals of the lightcurve fit is | 3.75 % |
| Best Comparison Star | #1 - [515, 253] |
| Optimal Aperture | 4.28 |
| Optimal Annulus | 11.15 |
| Transit Duration (day) | 0.0753 +/- 0.0053 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.205 ± 0.044 vs 0.16327 (deviation 0.95σ) |
| Duration fit vs published | 0.0753 d vs 0.0767 d (deviation 0.26σ) |
| Mid-time O–C | 1.5 min |
| Depth SNR | 4.7 |
| Residual scatter | 3.75 % |

## Light curve

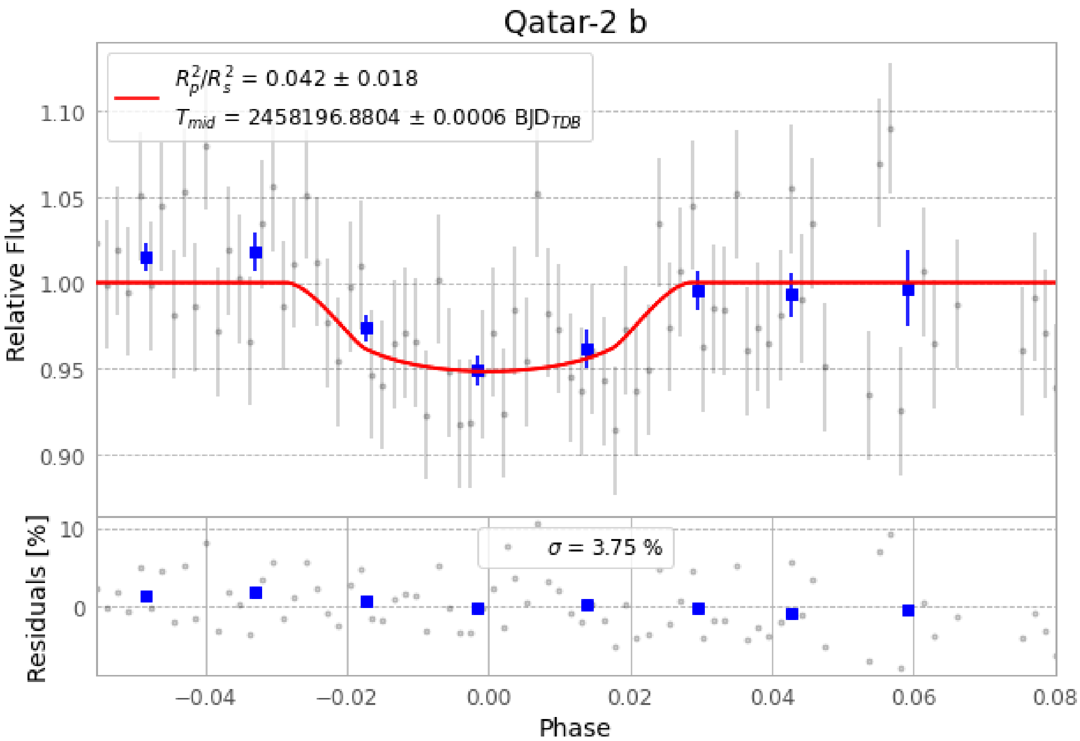


Figure 1 — FinalLightCurve_Qatar-2 b_2018-03-19.png (SHA-256 ec619e42a38e13b1…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [531, 186], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 90c5d071d110d5c7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

## Data provenance

88 FITS frames (58 MB), `Qatar-2180319071212.FITS` → `Qatar-2180319113312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180319.log (d8a3d477230f…), FinalLightCurve_Qatar-2 b_2018-03-19.png (ec619e42a38e…), FinalLightCurve_Qatar-2 b_2018-03-19.csv (eb7489c7d305…)

## Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)